A stochastic viscous step in the particle representation: plan

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Abstract

The deterministic viscous step in the particle representation (`changerep.tex`) holds each particle's energy fixed and moves momentum between particles so that the total particle entropy increases. Here we plan its stochastic version: random momentum transfers, accepted or rejected with the exact change of the particle entropy. The noise is fixed by the fluctuation–dissipation theorem through the quadratic form of the bond dissipative function, written as a sum of local pieces, one per bond and one per particle. This note records the plan and the questions to settle before coding.

I. THE STATE AND THE ENTROPY

During the viscous step the positions are fixed. Each particle has a fixed volume $V_a = s_a/S_a$ and a fixed energy, which with the gradient correction is $e'_a = e_a - g_a$ (`changerep.tex`, Appendix A). The momenta $\boldsymbol{m}_a$ move. The stochastic step maximizes the particle entropy subject to constraints on each particle's energy and on the energy of its long-wavelength gradient. During a sweep we therefore hold both e'_a and the gradient energy g_a fixed, so each particle's energy $e_a = e'_a + g_a$ is fixed, and the total particle entropy is

$$\mathbb{S}[\boldsymbol{m}] = \sum_a S_a\big(e_a,\ \boldsymbol{m}_a\big), \qquad \frac{\partial \mathbb{S}}{\partial \boldsymbol{m}_a} = -\beta_a, \tag{1}$$

local in each particle. The gradient energy is updated only at the end, in the back map $s'_a = S_a(e'_a + g'_a, \boldsymbol{m}'_a)$ of `changerep.tex`, as in the deterministic step.

The stochastic step should leave the distribution $e^{\mathbb{S}}$ at fixed energies and total momentum invariant. In equilibrium every particle-scale momentum mode then carries its thermal share, $\langle|\boldsymbol{m}_{\boldsymbol{k}}|^2\rangle \propto wTV$, at all k. This is the point of the change of representation: with the smoothed b the short-wavelength modes are damped by R^2 while their noise is not, and they run away.

II. A STRESS PROPOSAL

The stochastic step is a Metropolis update of the momenta. A momentum change $\delta\boldsymbol{m}$, conserving the total momentum, is drawn from a symmetric Gaussian with covariance Σ and accepted with a probability that depends on the change of the particle entropy $\Delta\mathbb{S}$. For small proposals the mean change is

$$\langle \Delta\boldsymbol{m} \rangle \simeq \frac{1}{2}\,\Sigma\,\frac{\partial \mathbb{S}}{\partial \boldsymbol{m}} = -\frac{1}{2}\,\Sigma\,\beta. \tag{2}$$

The deterministic step is $\Delta\boldsymbol{m} = -\Delta t\,\mathcal{K}\beta$, with $\mathcal{K}$ the quadratic form of the dissipative function, so the two agree for

$$\Sigma = 2\Delta t\,\mathcal{K}, \tag{3}$$

the fluctuation–dissipation theorem. The dissipative function is a sum of local pieces, $K = \sum_a K_a$, each depending only on the star of a, so the noise can be built one particle at a time. We build it from a random stress, start at rest, and check its mean traction.

a. The move. For particle a:

1. *Draw a random stress* ξ_a^{ij}, symmetric, with zero mean and variance

$$\langle \xi_a^{ij} \xi_a^{lm} \rangle = \frac{2\Delta t\,(T_{\rm rel})_a}{V_a}\,\kappa_a^{ijlm}, \tag{4}$$

where $(T_{\rm rel})_a$ is the rest-frame temperature of particle a. Our normal temperature is $T_a = \partial H/\partial s_a = (T_{\rm rel})_a/\gamma_a$, with $\beta_a = \boldsymbol{v}_a/T_a$; the rest-frame temperature appears only with the viscous coefficients, $T_{\rm rel}\kappa$, $T_{\rm rel}\eta$, $T_{\rm rel}\zeta$, which fix the noise. At rest the two coincide. Further, κ^{ijlm} is the viscous tensor at rest,

$$\kappa^{ijlm} = \eta\Big(\delta^{il}\delta^{jm} + \delta^{im}\delta^{jl} - \frac{2}{d}\,\delta^{ij}\delta^{lm}\Big) + \zeta\,\delta^{ij}\delta^{lm}. \tag{5}$$

2. *Distribute it across the star.* Each face of a carries the traction

$$\mathbf{t}_{ab} = \xi_a \boldsymbol{A}_{ab}, \qquad \boldsymbol{A}_{ab} = V_a\, w_{ab}\, \boldsymbol{r}_{ab}, \tag{6}$$

with $\boldsymbol{A}_{ab}$ the face of `bond_dissipation.tex`. Particle b receives $\mathbf{t}_{ab}$ and a gives up their sum:

$$\delta\boldsymbol{m}_b = \mathbf{t}_{ab}, \qquad \delta\boldsymbol{m}_a = -\sum_{b\neq a} \mathbf{t}_{ab}. \tag{7}$$

Momentum is conserved. By the closure of the faces the virial of the tractions is the stress itself,

$$\sum_{b\neq a} \mathbf{t}_{ab}\, \boldsymbol{r}_{ab}^{T} = V_a\, \xi_a, \tag{8}$$

which is symmetric, so the faces exert no net torque. This is the angular-momentum constraint.

3. *Ask the star.* Accept or reject the whole proposal by the consensus of its bonds, described next. On rejection the star is left untouched.

Each particle in turn does the same, so particle b, with its own stress ξ_b of variance κ_b, gives the traction $\mathbf{t}_{ba}$ to a.

b. The consensus. The proposal changes the entropy of a and of every member of its star,

$$\Delta\mathbb{S} = \Delta S_a + \sum_{b\neq a} \Delta S_b. \tag{9}$$

We write this as a sum over the bonds,

$$\Delta\mathbb{S} = \sum_{b\neq a} \Delta S_{ab}, \tag{10}$$

with one term per bond, specified below. A single vote would accept the proposal with probability $\min(1, e^{\Delta\mathbb{S}})$. Instead each bond votes, accepting with probability $\min(1, e^{\Delta S_{ab}})$, and the proposal is kept only if every bond accepts:

$$P(\xi_a) = \prod_{b\neq a} \min\left(1, e^{\Delta S_{ab}}\right). \tag{11}$$

If any bond rejects, the whole proposal is rejected. This is a factorized Metropolis filter. Like the single vote, it satisfies detailed balance with respect to $e^{\mathbb{S}}$, provided

1. the bond terms add up to $\Delta\mathbb{S}$, eq. (10);

2. each bond term flips its sign exactly under the reverse move, which starts from the new momenta and proposes $-\mathbf{t}_{ab}$ on every bond.

It rejects more often than the single vote: a bond that loses entropy can veto the move even when the others gain more.

c. Partitioning ΔS_a. Particle b receives only the traction $\mathbf{t}_{ab}$, so ΔS_b belongs to the bond ab. Particle a gives up the sum of all its tractions, so ΔS_a has to be divided among the bonds:

$$\Delta S_a = \sum_{b\neq a} \delta S_{ab}, \qquad \Delta S_{ab} = \delta S_{ab} + \Delta S_b. \tag{12}$$

Condition 1 then holds for any partition; condition 2 requires each δS_{ab} to flip its sign under the reverse move. (Counting ΔS_a in full on every bond would violate condition 1.) We take

$$\delta S_{ab} = \sigma_{ab} + \frac{R_a}{n_\mathrm{b}}, \tag{13}$$

where σ_{ab} is the part of ΔS_a due to $\mathbf{t}_{ab}$, R_a is the rest, and n_b is the number of bonds of a.

For σ_{ab}, with $\boldsymbol{m}_a$ and $\boldsymbol{m}'_a = \boldsymbol{m}_a + \delta\boldsymbol{m}_a$ the momenta of a before and after the move,

$$\sigma_{ab} = \frac{1}{2}\Big[S_a(\boldsymbol{m}_a - \mathbf{t}_{ab}) - S_a(\boldsymbol{m}_a)\Big] - \frac{1}{2}\Big[S_a(\boldsymbol{m}'_a + \mathbf{t}_{ab}) - S_a(\boldsymbol{m}'_a)\Big], \tag{14}$$

at fixed e_a, and the rest

$$R_a = \Delta S_a - \sum_{b\neq a} \sigma_{ab} \tag{15}$$

is split evenly. The reverse move exchanges $\boldsymbol{m}_a$ and $\boldsymbol{m}'_a$ and flips $\mathbf{t}_{ab}$, so it flips the sign of each σ_{ab} exactly, and of R_a: condition 2 holds. To leading order $\sigma_{ab} \simeq \beta_a \cdot \mathbf{t}_{ab}$, R_a is of third order, and

$$\Delta S_{ab} \simeq -\big(\beta_b - \beta_a\big) \cdot \mathbf{t}_{ab}\,: \tag{16}$$

each factor is the entropy produced by the bond itself. It depends only on the difference $\beta_b - \beta_a$, so it vanishes for a uniformly moving fluid, as the dissipation does.

For an entropy quadratic in the momentum the second-order terms cancel in eq. (14), and

$$\sigma_{ab} = \beta_a^\mathrm{mid} \cdot \mathbf{t}_{ab}, \qquad \Delta S_b = -\beta_b^\mathrm{mid} \cdot \mathbf{t}_{ab}, \tag{17}$$

where β^mid is evaluated at the midpoint of each particle's move: β_a^mid at the average of $\boldsymbol{m}_a$ and $\boldsymbol{m}'_a$, and β_b^mid at $\boldsymbol{m}_b + \mathbf{t}_{ab}/2$. The bond factor is then

$$\Delta S_{ab} = -\big(\beta_b^\mathrm{mid} - \beta_a^\mathrm{mid}\big) \cdot \mathbf{t}_{ab}, \tag{18}$$

the entropy produced by the bond, with the midpoint β at each end, exact up to third-order terms, which go into R_a.

The simpler even split, $\delta S_{ab} = \Delta S_a/n_\mathrm{b}$, also satisfies both conditions, but it is not invariant under a uniform shift of β: in a fluid moving with velocity $\boldsymbol{v}$, $\Delta S_b \simeq -\beta_b \cdot \mathbf{t}_{ab}$ contains $-(\boldsymbol{v}/T) \cdot \mathbf{t}_{ab}$, which is large and of random sign, and the bonds reject even though nothing dissipates. In a fluid moving at $v = 0.32$, the even split accepts 7% of the residual moves in 2D, and eq. (14) 47%, the same as at rest.

d. The mean traction. For a small proposal the entropy change is linear in the momentum changes, with $\partial S_c/\partial \boldsymbol{m}_c = -\beta_c$:

$$\Delta\mathbb{S} = -\sum_{b\neq a} \Delta\beta_{ab}\cdot\mathbf{t}_{ab} = -V_a\,\xi_a{:}G_a, \qquad G_a = \sum_{b\neq a} w_{ab}\,\Delta\beta_{ab}\,\boldsymbol{r}_{ab}^T, \tag{19}$$

where G_a is the fitted gradient of β at a (`bond_dissipation.tex`, mean-field form) and $\Delta\beta_{ab} = \beta_b - \beta_a$. The proposal sees β only through G_a.

e. Mean of the accepted stress. Each factor of eq. (11) is linear in the proposal to leading order,

$$\Delta S_{ab} \simeq g_{ab}{:}\xi_a, \qquad \sum_{b\neq a} g_{ab} = \frac{\partial\Delta\mathbb{S}}{\partial\xi_a} = -V_a G_a, \tag{20}$$

and is of order $\sqrt{\Delta t}$. To the same order the acceptance probability is

$$P(\xi_a) \simeq 1 + \sum_{b\neq a} \min\big(0,\ g_{ab}{:}\xi_a\big). \tag{21}$$

The proposal has zero mean, so the mean of the accepted stress is

$$\langle \xi_a\, P(\xi_a)\rangle = \sum_{b\neq a} \langle \xi_a\ \min\big(0,\ x_b\big)\rangle, \qquad x_b = g_{ab}{:}\xi_a. \tag{22}$$

Write $C = \langle \xi_a\xi_a\rangle$ for the covariance eq. (4). For a Gaussian ξ_a the part of ξ_a correlated with x_b is $C{:}g_{ab}\,x_b/\langle x_b^2\rangle$, and the rest is independent of x_b, so

$$\langle \xi_a\ \min(0,x_b)\rangle = \frac{C{:}g_{ab}}{\langle x_b^2\rangle}\,\langle x_b \min(0,x_b)\rangle = \frac{1}{2}\,C{:}g_{ab}, \tag{23}$$

since $x_b\min(0,x_b)$ is x_b^2 for half of the Gaussian and zero otherwise. Summing over the bonds,

$$\langle \xi_a P(\xi_a)\rangle = \frac{1}{2}\,C{:}\frac{\partial\Delta\mathbb{S}}{\partial\xi_a} = -\Delta t\,(T_{\rm rel})_a\,\kappa_a{:}G_a = \Delta t\,\Pi_a(G_a), \tag{24}$$

the viscous stress of the fitted gradient. A single test on the whole star, $\min(1, e^{\Delta\mathbb{S}})$, gives the same mean: the factorization changes the rejection rate, not the drift. The mean traction is therefore

$$\langle \mathbf{t}_{ab}\rangle = \Delta t\,\Pi_a(G_a)\,\boldsymbol{A}_{ab}. \tag{25}$$

Compare the deterministic traction of the bond form,

$$\mathbf{t}_{ab}^{\rm det} = -2V_a(T_{\rm rel})_a\eta_a\,w_{ab}\,\mathcal{D}_{ab}\beta = \Pi_a(G_a)\,\boldsymbol{A}_{ab} - 2V_a(T_{\rm rel})_a\eta_a\,w_{ab}\,\boldsymbol{\delta}_{ab}, \tag{26}$$

where $\boldsymbol{\delta}_{ab} = \Delta\beta_{ab} - G_a\boldsymbol{r}_{ab}$ is the residual of the fit (`bond_dissipation.tex`, resolved plus residual).

- For a linear field the residual vanishes and eq. (25) is the deterministic traction: shear, bulk and rotation all come out right.
- For a general field eq. (25) is the traction of the *resolved* part of K only. The residual part, which damps the modes the fit cannot see, is missing. Those modes are then neither damped nor excited by the noise.

III. THE RESIDUAL MOVE

a. The missing part. The dissipative function splits exactly into a resolved and a residual part, with no cross term (`bond_dissipation.tex`):

$$K = K_{\rm res} + K_{\rm resid}, \qquad K_{\rm resid} = \sum_a V_a (T_{\rm rel})_a \eta_a \sum_{b\neq a} w_{ab}\, |\boldsymbol{\delta}_{ab}|^2. \tag{27}$$

The stress move of Sect. II is the noise of $K_{\rm res}$. The noise of $K_{\rm resid}$ is added on top of it, as a second, independent move. Since there is no cross term, the covariances add, and so do the mean tractions.

b. The move. For particle a:

1. *Draw a random vector on each bond*, independent from bond to bond, with zero mean and variance
$$\langle \boldsymbol{y}_{ab} \boldsymbol{y}_{ab}^T \rangle = 2\Delta t \cdot 2V_a (T_{\rm rel})_a \eta_a\, w_{ab}\, \mathbb{1}. \tag{28}$$
In 1D replace $2\eta_a$ by κ_a.

2. *Remove the part the fit sees.* Form the virial of the bond vectors,
$$Y_a = \sum_{b\neq a} \boldsymbol{y}_{ab}\, \boldsymbol{r}_{ab}^T, \tag{29}$$
and subtract it as a stress distributed through the faces. The residual traction is
$$\vec{t}_{ab} = \boldsymbol{y}_{ab} - \frac{Y_a}{V_a}\, \boldsymbol{A}_{ab}. \tag{30}$$
By the closure of the faces its virial vanishes,
$$\sum_{b\neq a} \vec{t}_{ab}\, \boldsymbol{r}_{ab}^T = Y_a - Y_a = 0, \tag{31}$$
so the residual tractions carry no stress and exert no torque.

3. *Kick and ask the bonds.* Particle b receives $\vec{t}_{ab}$ and a gives up their sum. Accept with the factorized filter eq. (11), with the bond factors eq. (12).

c. The mean traction. To leading order the entropy change of the star is

$$\Delta\mathbb{S} = -\sum_{b\neq a} \Delta\beta_{ab} \cdot \vec{t}_{ab} = -\sum_{b\neq a} \boldsymbol{y}_{ab} \cdot \boldsymbol{\delta}_{ab}, \tag{32}$$

because the subtracted stress removes $Y_a{:}G_a$ from the pair term. The argument of Sect. II gives

$$\langle \boldsymbol{y}_{ab} \rangle = -\Delta t \cdot 2V_a (T_{\rm rel})_a \eta_a\, w_{ab}\, \boldsymbol{\delta}_{ab}. \tag{33}$$

The residuals are orthogonal to the bonds,

$$\sum_{b\neq a} w_{ab}\,\boldsymbol{\delta}_{ab}\,\boldsymbol{r}_{ab}^{T} = 0, \tag{34}$$

so the mean of Y_a vanishes, and so does the mean of the subtracted stress. The mean residual traction is

$$\langle \vec{t}_{ab} \rangle = -\Delta t \cdot 2V_a(T_{\rm rel})_a \eta_a\, w_{ab}\,\boldsymbol{\delta}_{ab}. \tag{35}$$

Added to the mean stress traction eq. (25),

$$\langle \mathbf{t}_{ab} \rangle + \langle \vec{t}_{ab} \rangle = \Delta t \Big[\Pi_a(G_a)\,\boldsymbol{A}_{ab} - 2V_a(T_{\rm rel})_a \eta_a\, w_{ab}\,\boldsymbol{\delta}_{ab} \Big] = \Delta t\, \mathbf{t}_{ab}^{\rm det}, \tag{36}$$

the deterministic traction of the bond form, on every bond.

d. One vote or two. The two moves could also be drawn together and accepted by a single vote. Both are exact. We take two separate moves:

- The stress move stays as in Sect. II.
- Each vote sees less noise, so each is rejected less often.
- The residual move can be switched off, to check that it is what damps the modes the fit cannot see.

In 1D there is no rotation, and the residual move is what damps the zone-edge mode.

e. A cold residual. The residual move need not satisfy detailed balance. The modes it acts on are lattice modes, which we want damped but do not need to fluctuate. A one-parameter family keeps its mean traction and reduces its noise. For $\lambda \geq 1$:

- shrink the proposal covariance by $1/\lambda$;
- accept with $\min(1, e^{\lambda \Delta S_{ab}})$ in place of $\min(1, e^{\Delta S_{ab}})$ in eq. (11).

By the argument of Sect. II, the mean is multiplied by λ through the acceptance and divided by λ through the covariance, so the mean residual traction is unchanged. The noise is reduced by $1/\lambda$: the residual drives the modes toward the temperature T/λ.

- $\lambda = 1$ is detailed balance.
- $\lambda \to \infty$ is the deterministic residual step, done directly: the residual part of the force of `changerep.tex`, at fixed e_a, with no accept/reject.

The argument requires the exponent $\lambda \Delta S_{ab}$ to be small. It is of order $\sqrt{\lambda \Delta t}$, so the family holds for $\lambda \Delta t \ll 1$.

The reduction of the covariance is essential. Raising the exponent alone, or accepting only moves with $\Delta S > 0$, gives a mean kick of fixed size, of order $\sqrt{\Delta t}$, along the gradient, independent of the residual itself. That is a friction of fixed magnitude, not a viscosity, and it does not converge as $\Delta t \to 0$.

f. The effective temperature. A mode of wavenumber k receives noise from the stress move at temperature T and from the residual move at temperature T/λ, and is damped by both. Its steady state has

$$\frac{T_{\text{eff}}(k)}{T} = \frac{K_{\text{res}}(k) + K_{\text{resid}}(k)/\lambda}{K_{\text{res}}(k) + K_{\text{resid}}(k)}. \tag{37}$$

For a smooth mode $K_{\text{res}} \propto k^2$ and $K_{\text{resid}} \propto k^4 h^2$, so

$$\frac{T_{\text{eff}}(k)}{T} = 1 - O\big((kh)^2\big), \tag{38}$$

and the hydrodynamic modes fluctuate at the right temperature. At the zone edge the fit sees nothing, $K_{\text{res}} = 0$, and $T_{\text{eff}} = T/\lambda$. In between the crossover is smooth. This is a smooth regulator: the cutoff-dependent renormalization of the transport coefficients comes from the thermal modes up to the cutoff, now weighted by eq. (37). The counterterm changes, but it can be computed in the same way. The total entropy is, to quadratic order, a constant minus the kinetic energy over T, so the steady state does not heat.

IV. THE SWEEP

a. Independent moves. A move of particle a kicks only its star, the particles within $2h$, and its $\Delta\mathbb{S}$ involves only those particles, since $\mathbb{S}$ is a sum of single-particle entropies at fixed e_a and V_a. Two moves therefore commute exactly when their centres are more than $4h$ apart.

b. Checkerboard. The cells of the cell list have side at least $2h$. Color them by their indices modulo 3 in each direction: 9 colors in 2D, 3 in 1D. Cells of the same color are then at least two cells, i.e. $4h$, apart, and the moves in them are independent. If the number of cells in a direction is not a multiple of 3, the periodic wrap breaks the pattern; either choose it a multiple of 3 or give the leftover column its own color. One sweep:

1. Visit the colors in a random order.

2. Within a color, treat every active cell independently (in parallel).

3. Inside a cell, visit its particles in a random order, or pick particles at random. Each particle makes its stress move, using the current momenta.

The residual moves are made in a second, separate sweep of the same form, after the stress sweep is complete. The order of the two sweeps can be alternated from step to step. Each move satisfies detailed balance on its own, and moves of the same color act on disjoint sets, so each sweep leaves $e^{\mathbb{S}}$ stationary, and so does their composition. The mean drifts of the two sweeps add to the deterministic step at leading order in Δt. Nothing is double counted: every accepted move sees the momenta left by the previous ones. The random color order makes the sweep reversible on average.

c. Details.

- *Frozen coefficients.* The variances depend on $(T_{\mathrm{rel}})_a$, w_{ab} and the fit. They are computed at the start of the viscous step and held fixed, so that each proposal is a fixed symmetric Gaussian and the Metropolis test is exact. Recomputing them from the current momenta would make the proposal depend on the state and require a Hastings correction.

- *Random grid shift.* Before each sweep the grid of cells is shifted by a random vector $\boldsymbol{u}$, uniform within one cell, and the particles are re-binned by $\boldsymbol{x}_a + \boldsymbol{u}$ (modulo the box). Only the partition into cells changes; the stars are frozen and do not depend on it. The cells keep their size, so they still tile the box and are at least $2h$ wide. The re-binning is a single pass over the particles. With fixed positions the shift is not needed for correctness; it removes any imprint of the cell boundaries.

- *Random streams.* One stream per cell or per particle, not per thread, so the results do not depend on the number of threads.

- *Mean drift.* Earlier moves shift the momenta of a star by zero-mean noise of order $\sqrt{\Delta t}$. At leading order in Δt the mean drift is still the deterministic step.

d. Pseudo-code. As implemented in `src/stochrep.jl` (`stochrep_step!`). The colors along a direction of n cells are $0, 1, 2, 0, 1, 2, \ldots$ for the first $3\lfloor n/3 \rfloor$ cells and $3, 4$ for the one or two leftover cells; the color of a cell is the pair of its two colors.

```
stochrep_step!(particles, dt; lambda):
    # frozen for the step (or for the whole run on a frozen lattice)
    for each particle a:
        V[a], e[a]                          # particle volume and energy
        star[a] = bonds b within 2h:  r_ab, w_ab, c_ab
        Trel_a, eta_a, zeta_eff_a              # rest frame
    m[a] = current momenta;  S[a] = S_a(e[a], m[a])
    for kind in (stress, residual)          # order alternates step to step
        if kind == residual and lambda == Inf
            deterministic residual step;  continue
        u = random vector within one cell;  re-bin particles by x + u
        for color in random_permutation(colors)
            parallel for cell in cells_of(color)
                for a in random_permutation(particles_in(cell))
                    move(a, kind)
    write back m[a] and s[a] = S[a]

move(a, kind):                              # random numbers from rngs[a]
    if kind == stress
        xi = random symmetric stress, <xi xi> = 2 dt Trel_a kappa_a / V_a
        delta[b] = xi A_ab,                 A_ab = V_a w_ab r_ab
        lam = 1
    else
        y[b] = random vector, <y y^T> = 2 dt c_ab / lambda
```

```
    Y = sum_b y[b] r_ab^T
    delta[b] = y[b] - (Y / V_a) A_ab
    lam = lambda
delta_a = - sum_b delta[b]
m'_a = m[a] + delta_a
dSa  = S_a(e[a], m'_a) - S[a]
for b:  sigma[b] = (S_a(m[a] - delta[b]) - S_a(m[a]))/2
                 - (S_a(m'_a + delta[b]) - S_a(m'_a))/2
R    = dSa - sum_b sigma[b]
logP = sum_b min(0, lam (S_b(e[b], m[b] + delta[b]) - S[b]
                         + sigma[b] + R/n_a))
if log(uniform) < logP:                  # all or nothing
    m[b] += delta[b] for every bond;  m[a] += delta_a;  update S
```

A single vote of the star, $\log P = \min(0, \lambda\,\Delta\mathbb{S})$, is also available (`vote=:star`); it is exact as well and rejects less often. The even split is `vote=:even`.

V. TESTS ON A FROZEN LATTICE

a. Setup. A lattice of particles with equal particle energies e_a and zero momenta. Positions, volumes V_a and energies e_a are fixed; only the viscous step acts. The conformal equation of state with $\eta/s = 0.2$, $\zeta = 0$.

- 1D: $h = 0.05$, $h/a = 4$, $L = 1$ (80 particles), 15 bonds per star.
- 2D: hexagonal lattice, $h = 0.05$, $h/a = 2.5$, a box of side 0.6 (1020 particles), 88 bonds per star.

The entropy per particle is $s_a = 500$ (1D) and 346 (2D), so the thermal velocities are small, about 0.05. The time step is a quarter of the default viscous half step of `changerep.tex`. Scripts: `examples/stochrep_thermalize.jl`, `examples/stochrep_smallk.jl`.

b. The equilibrium value. For one particle at fixed e_a, $e^{S_a(e_a,\boldsymbol{m})}$ gives each momentum component the variance $V_a wT$ to leading order. A direct numerical integral gives $0.993\,V_a wT$ at $s_a = 500$; the difference is a relativistic correction. This is the reference value below.

c. Detailed balance. With $\lambda = 1$ the sweep must reproduce it exactly, for any Δt. Starting from a thermal draw, the time-averaged $\langle m^2\rangle/(V_a wT)$ in 1D is 0.990 ± 0.004 with the bond vote, 0.991 ± 0.003 with the even split and 0.996 ± 0.004 with the single vote, and 0.992 ± 0.004 with the grid shift.

d. The residual damping. For a single small mode, one deterministic residual step reduces the amplitude at exactly the rate $2K_{\rm resid}(k)/(V_a wT)$ predicted by the star (ratio 1.0000 for every k).

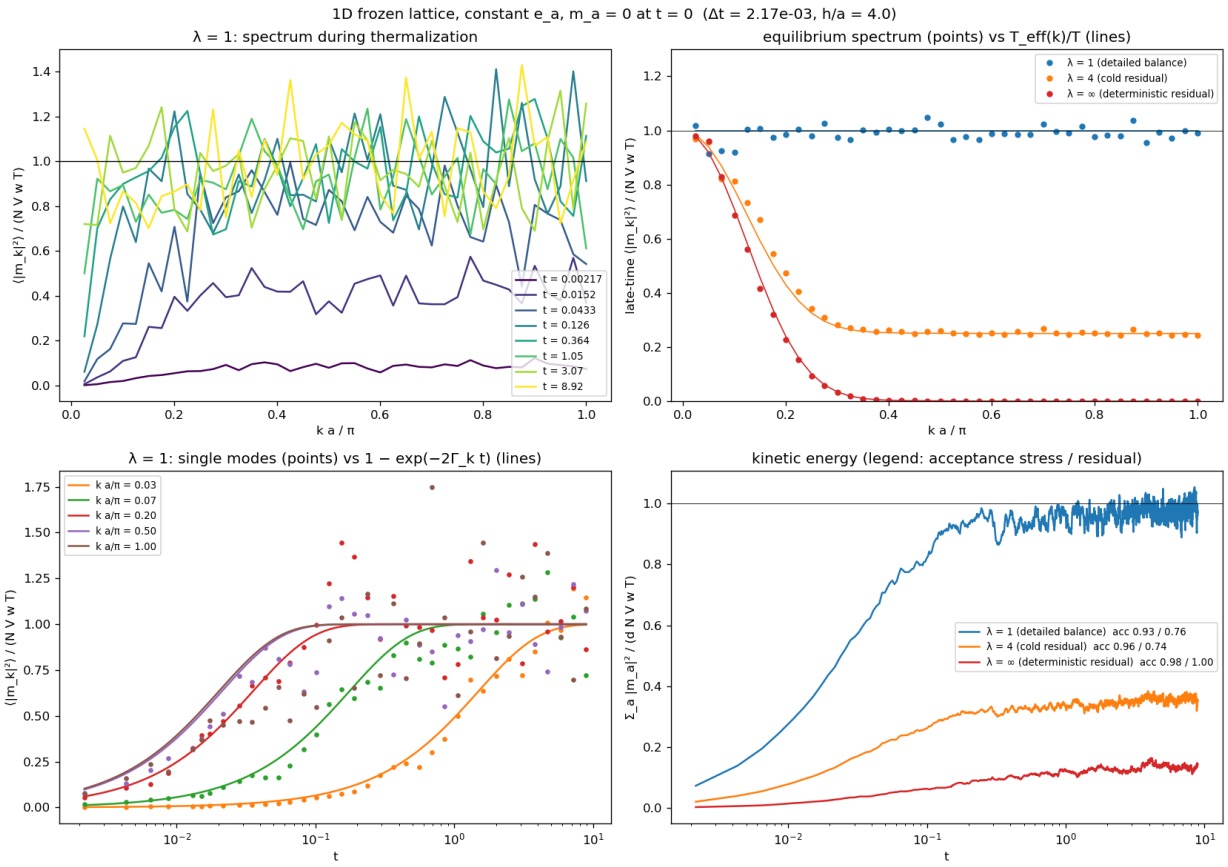


FIG. 1. 1D frozen lattice, thermalization from zero momenta. Top left: the spectrum at several times ($\lambda = 1$). Top right: the late-time spectrum for $\lambda = 1, 4, \infty$ (points) and $T_{\rm eff}(k)/T$ (lines). Bottom left: single modes (points) and $1 - e^{-2\Gamma_k t}$ (lines). Bottom right: kinetic energy, and the acceptance of the stress and residual moves.

e. Thermalization. Fig. 1 and Fig. 2. From zero momenta the spectrum fills in from the zone edge, where the damping is fastest, and each mode approaches its equilibrium as $1 - e^{-2\Gamma_k t}$, with Γ_k the damping rate of the bond form. The late-time spectrum is flat for $\lambda = 1$ and follows $T_{\rm eff}(k)/T$, eq. (37), for $\lambda = 4$ and $\lambda = \infty$, with no fitted parameter. In 2D this holds for both polarizations. The modes damped mostly by the residual approach their equilibrium somewhat more slowly than predicted, because of the rejections.

f. Small k. Fig. 3: a 1D lattice of length 2 (160 particles), started from a thermal draw, 64 runs. Every mode shown except the lowest relaxes at least ten times during the run; the lowest only 2.5 times, and its error is large (a run eight times longer gives 0.965 ± 0.032 there, against 0.995). For $\lambda = \infty$ the deficit $1 - T_{\rm eff}/T$ vanishes as k^2, and the measurement follows it. Close to the prediction there remains a small deficit that is a time-step error: relative to $T_{\rm eff}$, at $ka/\pi = 0.1$–0.2,

Δt / default	bond vote	single vote
1/4	−3%	−3%
1/16	−1.5%	−0.5%

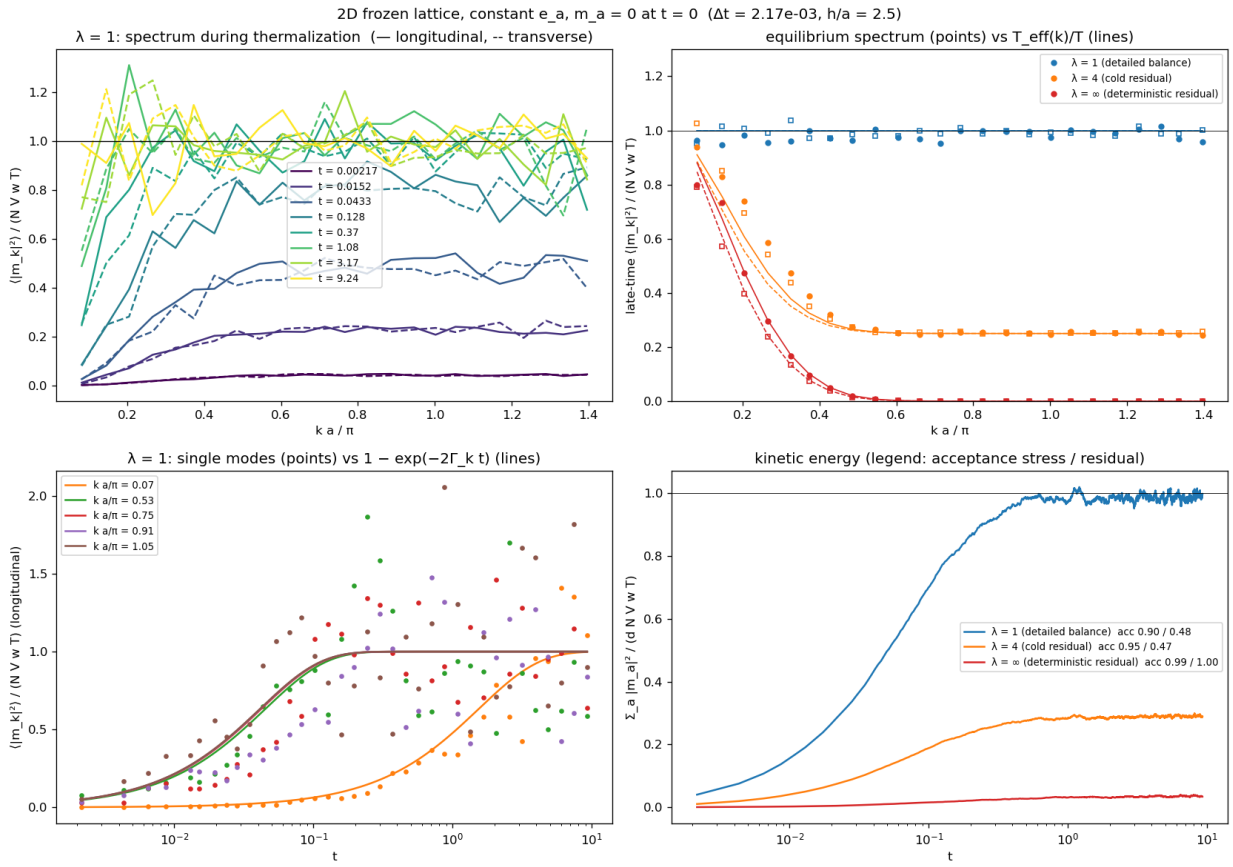


FIG. 2. As Fig. 1, in 2D. Solid lines and filled points: longitudinal; dashed lines and open points: transverse.

It does not depend on the vote, so it comes from the explicit residual step, not from the rejections.

g. Acceptance. Rejections fall as $\sqrt{\Delta t}$. The residual move, with 88 bonds in 2D, is rejected most:

residual move	bond vote, eq. (14)	even split	single vote
1D, at rest	0.75	0.79	0.89
1D, moving at $v = 0.32$	0.73	0.18	0.89
2D, at rest	0.48	0.58	0.88
2D, moving at $v = 0.32$	0.47	0.07	0.88

In 2D, at the full default step, the bond vote accepts 0.35 of the residual moves and the single vote 0.77; at 1/16 of it, 0.76 and 0.94. These tables were measured without the grid shift.

h. Still to do.

1. The proposal coefficients are evaluated in the rest frame; they need the comoving forms of the deterministic step for a moving fluid.

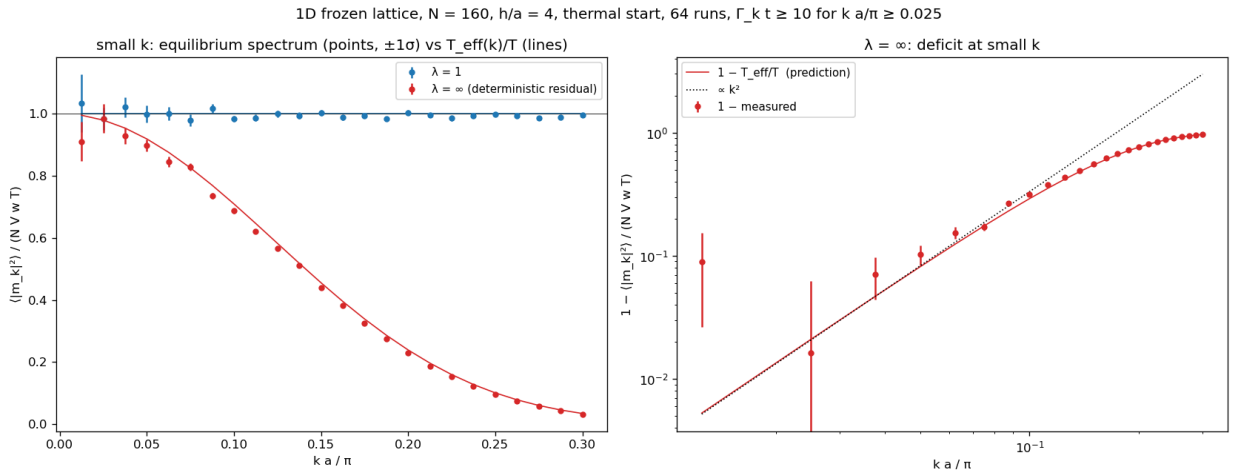


FIG. 3. 1D, small k. Left: the equilibrium spectrum ($\pm 1\sigma$) for $\lambda = 1$ and $\lambda = \infty$, with $T_{\rm eff}(k)/T$ (lines). Right: the deficit $1 - \langle |m_k|^2 \rangle / (NV_a wT)$ for $\lambda = \infty$, and the prediction, $\propto k^2$.

2. *Energy.* Follow H and the temperature over long runs.

3. *Hydrodynamics.* With the Hamiltonian step on: the equilibrium spectrum of a uniform fluid, and ensemble-averaged sound and shear decay.

VI. IS THE RESIDUAL MOVE NEEDED?

With the stress move alone a mode is damped, and receives noise, only through the fitted gradient, at the rate $R(k)\Gamma(k)$, where $R(k) = K_{\rm res}(k)/K(k)$ is the fraction of the bond dissipation that the fit sees. R falls from one for $kh \ll 1$ to zero beyond $kh \simeq 4$, so most lattice modes are invisible to the stress move. With the Hamiltonian step on, these modes are heated by the ideal SPH force and run away unless the residual move damps them. The runs that show this, and the first long runs with the Hamiltonian step, are documented in `fullruns.tex`.

VII. A MOVING FLUID

The construction carries over to a fluid moving with velocity $\boldsymbol{v}$, with the same substitutions as in the deterministic bond form (`bond_dissipation.tex`, the moving fluid): bonds are measured in the rest frame of the particle, the metric δ is replaced by the comoving metric ℓ, and the scalar part of the strain is removed along the projector P. The consensus, the partition of ΔS_a and the sweep are unchanged.

a. Ingredients. For particle a, with velocity $\boldsymbol{v}_a$, $\gamma_a = 1/\sqrt{1 - v_a^2}$, temperature $T_a = \partial H/\partial s_a$ and rest-frame temperature $(T_{\rm rel})_a = \gamma_a T_a$:

- the thermal vector is $\beta_a = \boldsymbol{v}_a/T_a = \gamma_a \boldsymbol{v}_a/(T_{\rm rel})_a$, and still

$$\left.\frac{\partial S_a}{\partial \boldsymbol{m}_a}\right|_{e_a} = -\beta_a; \tag{39}$$

- the comoving metric and the scalar projector,

$$\ell^{ij} = \delta^{ij} - v^i v^j, \qquad P_{ij} = -c_s^2 u_i u_j + \frac{1}{d}\big(\delta_{ij} + u_i u_j\big), \qquad \langle P\ell\rangle \equiv P_{ij}\ell^{ij} = 1 - c_s^2 v^2, \tag{40}$$

  with $u_i = \gamma v_i$;

- bond weights evaluated at the rest-frame bond length, normalized so that the bond moment is the comoving metric,

$$\sum_{b\neq a} w_{ab}\, \boldsymbol{r}_{ab}\boldsymbol{r}_{ab}^T = \ell_a; \tag{41}$$

- the face covector

$$A_{ab\,k} = V_a\, w_{ab}\, (\ell_a^{-1}\boldsymbol{r}_{ab})_k, \qquad \sum_{b\neq a} A_{ab\,k}\, r_{ab}^j = V_a\, \delta_k{}^j; \tag{42}$$

- the fit, now

$$G_a = \sum_{b\neq a} w_{ab}\, \Delta\beta_{ab}\, \boldsymbol{r}_{ab}^T\, \ell_a^{-1}, \qquad G_{a,ik} = \sum_{b\neq a} \frac{1}{V_a}\, (\Delta\beta_{ab})_i\, A_{ab\,k}. \tag{43}$$

b. Which velocity. The comoving metric, the projector, the bond lengths and $\kappa(\boldsymbol{v})$ are coefficients of the coarse-grained flow. They are evaluated with the coarse-grained particle velocity

$$\boldsymbol{v}_a \to \dot{\boldsymbol{x}}_a = \frac{\partial H}{\partial \boldsymbol{m}_a}, \tag{44}$$

the velocity with which the particles move, which varies slowly. The particle's own velocity, from $(e_a, \boldsymbol{m}_a)$, fluctuates thermally and is not used here; it enters only through β_a in the entropy. The coefficients are frozen at the start of the step, so the proposals are fixed symmetric Gaussians as before, and $e^{\mathbb{S}}$ is stationary whatever velocity is used; the choice affects only the coefficients, at order h^2.

c. The stress move. The resolved part of the dissipative function is

$$K_{\mathrm{res},a} = \frac{1}{2} V_a (T_{\mathrm{rel}})_a\, \kappa_a^{ijmn}(\boldsymbol{v}_a)\, G_{a,ij} G_{a,mn} = V_a (T_{\mathrm{rel}})_a \Big[\eta_a\, \mathrm{tr}\,\big(\ell_a \mathring{S}_a \ell_a \mathring{S}_a\big) + \frac{1}{2}\zeta_a \theta_a^2\Big], \tag{45}$$

with $\mathring{S} = S - \theta P$, S the symmetric part of G and $\theta = \ell^{ij} S_{ij}/\langle P\ell\rangle$. Draw the stress with

$$\langle \xi_a^{ij} \xi_a^{mn}\rangle = \frac{2\Delta t\, (T_{\mathrm{rel}})_a}{V_a}\, \kappa_a^{ijmn}(\boldsymbol{v}_a), \tag{46}$$

and distribute it through the faces,

$$\mathsf{t}_{ab}^i = \xi_a^{ik} A_{ab\,k}. \tag{47}$$

By eq. (42) the virial is again the stress,

$$\sum_{b\neq a} \mathsf{t}^i_{ab}\, r^j_{ab} = V_a\, \xi^{ij}_a, \tag{48}$$

which is symmetric: no torque. The linear entropy change is, by eq. (43),

$$\Delta\mathbb{S} = -\sum_{b\neq a} (\Delta\beta_{ab})_i\, \mathsf{t}^i_{ab} = -V_a\, \xi^{ik}_a G_{a,ik}, \tag{49}$$

exactly as at rest, and the argument of Sect. II gives

$$\langle \xi^{ij}_a \rangle = -\Delta t\, (T_{\rm rel})_a\, \kappa^{ijmn}_a G_{a,mn} = \Delta t\, \Pi^{ij}_a(G_a), \qquad \langle \mathsf{t}^i_{ab} \rangle = \Delta t\, \Pi^{ik}_a(G_a)\, A_{ab\,k}, \tag{50}$$

the traction of the resolved part of the moving bond form.

d. Drawing the stress. Write the quadratic form in whitened variables. With $\hat e_1 = (\hat x\hat x - \hat y\hat y)/\sqrt{2}$ and $\hat e_2 = (\hat x\hat y + \hat y\hat x)/\sqrt{2}$ an orthonormal basis of traceless symmetric matrices, define

$$E_k = \ell^{1/2}\hat e_k\, \ell^{1/2} - \frac{P_{ij}(\ell^{1/2}\hat e_k \ell^{1/2})^{ij}}{\langle P\ell\rangle}\, \ell, \qquad k = 1, 2. \tag{51}$$

Then, with z_1, z_2, z_3 independent unit Gaussians,

$$\xi_a = \sqrt{\frac{2\Delta t\, (T_{\rm rel})_a}{V_a}} \Big[\sqrt{2\eta_a}\, \big(z_1 E_1 + z_2 E_2\big) + \sqrt{\zeta_a}\, z_3\, \frac{\ell}{\langle P\ell\rangle} \Big] \tag{52}$$

has the covariance above. At rest $E_k = \hat e_k$ and $\ell = \mathbb{1}$, and this is the draw of Sect. II. (With unresolved dimensions, $\zeta \to \zeta_{\rm eff}$.)

e. The residual move. The residual part is

$$K_{{\rm resid},a} = V_a (T_{\rm rel})_a \eta_a \sum_{b\neq a} w_{ab}\, \boldsymbol{\delta}_{ab} \cdot \ell_a \cdot \boldsymbol{\delta}_{ab}, \qquad \boldsymbol{\delta}_{ab} = \Delta\beta_{ab} - G_a \boldsymbol{r}_{ab}. \tag{53}$$

Draw on each bond

$$\boldsymbol{y}_{ab} = \sqrt{2\Delta t\, c_{ab}/\lambda}\; \ell_a^{1/2} \boldsymbol{z}_{ab}, \qquad c_{ab} = 2V_a (T_{\rm rel})_a \eta_a\, w_{ab}, \tag{54}$$

with $\boldsymbol{z}_{ab}$ a vector of independent unit Gaussians, so $\langle \boldsymbol{y}_{ab} \boldsymbol{y}^T_{ab} \rangle = 2\Delta t\, c_{ab}\, \ell_a/\lambda$. Remove the part the fit sees, now through the face covector:

$$Y_a = \sum_{b\neq a} \boldsymbol{y}_{ab}\, \boldsymbol{r}^T_{ab}, \qquad \vec t^{\,i}_{ab} = y^i_{ab} - \frac{1}{V_a}\, Y^{ik}_a A_{ab\,k}. \tag{55}$$

By eq. (42) its virial vanishes, $\sum_b \vec t^{\,i}_{ab} r^j_{ab} = Y^{ij}_a - Y^{ij}_a = 0$. The linear entropy change is $-\sum_b \boldsymbol{y}_{ab} \cdot \boldsymbol{\delta}_{ab}$, and since the residuals are still orthogonal to the bonds (the orthogonality involves no metric), the mean of Y_a vanishes and

$$\langle \vec t^{\,i}_{ab} \rangle = -\Delta t \cdot 2V_a (T_{\rm rel})_a \eta_a\, w_{ab}\, \ell^{ij}_a (\delta_{ab})_j, \tag{56}$$

the residual part of the moving bond traction. The two moves together give the deterministic traction $-2V_a(T_{\rm rel})_a \eta_a w_{ab}\, \ell_a\, \mathcal{D}_{ab}\beta$ on every bond, plus the mean-field force of the bulk and dilation terms, as at rest.

f. The consensus. Unchanged. The bond factor is still, to leading order, $-(\beta_b-\beta_a)\cdot\mathbf{t}_{ab}$, so it is not affected by the background flow; this is why the partition eq. (14) was chosen (Sect. V: in a fluid moving at $v = 0.32$ the acceptance is the same as at rest).

g. The equilibrium. The stationary distribution is still $e^{\mathbb{S}}$ at fixed e_a. Around a moving state the momentum fluctuations are no longer isotropic. To quadratic order their covariance per particle is the inverse of minus the Hessian of S_a at fixed e_a,

$$\langle \delta m^i\, \delta m^j\rangle = V_a \left[\left(\chi^{-1}\right)_{ij}\right]^{-1}, \qquad \chi^{-1}_{\mu\nu} = \frac{\partial \beta_\mu}{\partial T^{0\nu}}, \tag{57}$$

the spatial block of the inverse susceptibility of `theory.tex`, inverted. It is not the spatial block of χ itself: holding e_a fixed conditions on the energy. From `theory.tex`,

$$\chi^{-1}_{ij} = \frac{1}{(e+p)\gamma\, T_{\rm rel}}\Big[\frac{4c_s^2}{1-c_s^2v^2}\, u_iu_j + \delta_{ij} + u_iu_j\Big]. \tag{58}$$

For $v = 0.5$ along x and $c_s^2 = 1/3$ this gives $C_{xx} = 0.733\, V_a wT$ along the flow and $C_{yy} = 1.333\, V_a wT$ across it; a numerical Hessian of S_a agrees to four digits.

Beyond quadratic order, the single-particle distribution in a large system at fixed total momentum is $e^{S_a(e_a,\boldsymbol{m}_0+\delta\boldsymbol{m})+\boldsymbol{\lambda}\cdot\delta\boldsymbol{m}}$, with the multiplier $\boldsymbol{\lambda}$ of the total momentum chosen so that $\langle\delta\boldsymbol{m}\rangle = 0$ ($\boldsymbol{\lambda} \simeq \beta_0$; setting it equal to β_0 shifts the mean by 0.09σ along the flow and the covariance by 1.7%). At $s_a = 346$ its covariance is $C_{xx} = 0.722\, V_a wT$ and $C_{yy} = 1.316\, V_a wT$, within 1.5% of eq. (57).

h. Test. Fig. 4: the 2D frozen hexagonal lattice of Sect. V, moving at $v = 0.5$ along x (and at rest, for comparison), stochastic step only, $\lambda = 1$, starting from $\delta\boldsymbol{m} = 0$. Eight runs.

- The stress noise reproduces $\kappa^{ijmn}(\boldsymbol{v})$ of the code to machine precision, at $v = 0$, 0.58 and 0.79, with and without bulk viscosity.
- The late-time spectrum, normalized by the exact covariance, is flat in k for both polarizations. Averaged over particles, $\langle\delta m_x^2\rangle/C_{xx} = 0.996$ and $\langle\delta m_y^2\rangle/C_{yy} = 0.999$ at $v = 0.5$; a separate run from a thermal start gives $\langle\delta m_x^2\rangle/C_{xx} = 1.000 \pm 0.002$, with either vote.
- The acceptance is the same as at rest, and the total momentum is conserved to 10^{-14}.

i. Implementation. In 2D the code uses the comoving forms above: comoving weights and ℓ for each star, the face covector eq. (42) in both moves, and the draws above. The velocity is $\dot{\boldsymbol{x}}_a$ (`pa.dotx`), in the stochastic step and in the deterministic bond force alike, so that the mean of the one equals the other. The coefficients use $(T_{\rm rel})_a = \gamma_a\, \partial H/\partial s_a$; in the code `pa.T` is $\partial H/\partial s_a = (T_{\rm rel})_a/\gamma_a$. In 1D the rest-frame forms are used.

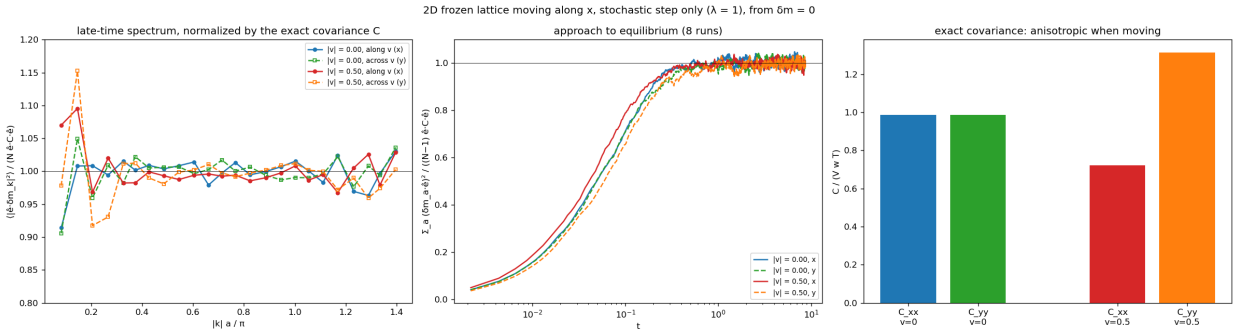


FIG. 4. 2D frozen lattice moving along x at $v = 0.5$, and at rest; stochastic step only, $\lambda = 1$, from $\delta\boldsymbol{m} = 0$. Left: late-time spectrum of the momentum fluctuations along and across the flow, normalized by the exact covariance. Middle: approach to equilibrium. Right: the exact covariance, anisotropic when the lattice moves.